

Submission A solution to Problem 10

Problem statement and interpretation

Irreducible means not a non-trivial graph join; equivalently the complement graph Γ^c is connected. All traces are faithful normal traces. For $U \subset V$, M_U denotes the graph product over the induced subgraph and $E_U : M \rightarrow M_U$ its canonical trace-preserving expectation.

We use the notation of Ding–Kunnawalkam Elayavalli–Peterson [1]. For a finite algebra N , put $\mathcal{A}_N = \mathbb{B}(L^2 N)$ and

$$\mathcal{A}_N = (A_N^{\sharp_J})^*,$$

where $\sharp_J$ means simultaneous normality for the N - and JNJ -bimodule structures. If X is an N -boundary piece, $K_X(N)$ is DKEP's compact hereditary algebra and $q_X \in \mathcal{A}_N$ is the support of $(K_X(N)^{\sharp_J})^*$. For $X = \mathbb{K}(L^2 N)$ write q_0 . Set

$$\tilde{S}_X(N) = \{T \in \mathcal{A}_N : [T, a] \in q_X \mathcal{A}_N q_X \text{ for all } a \in JNJ\}.$$

For a separable finite factor, DKEP Lemma 8.5 says that proper proximality relative to X is equivalent to the absence of an N -central state on $\tilde{S}_X(N)$ whose restriction to N is the trace. Equivalently one may rule out N -central states whose restriction is normal, since a normal central state on a finite factor is the trace after normalization.

For a bounded vector ξ in an M - N correspondence, T_ξ denotes DKEP's coefficient operator $L^2 M \rightarrow L^2 N$; if η is bounded in an N - P correspondence, then

$$T_{\xi \otimes_N \eta} = T_\eta T_\xi, \quad T_\xi = T_\xi^*. \tag{1}$$

Boundary supports and coefficients

We use DKEP's ordinary-bidual realization of the normal bidual. Thus $\mathcal{A}_N$ is viewed as a von Neumann subalgebra of A_N^{**} ; its unit is denoted by p_N . Products of support projections are computed inside the honest von Neumann algebra A_N^{**} , not by multiplying the non-multiplicative canonical images of concrete operators.

Lemma 1 (ordinary support bookkeeping). *Let $Y \subset A_N$ be a hereditary C^* -subalgebra invariant under the multiplier actions of N and JNJ , and assume that the induced representations $N, JNJ \rightarrow M(Y)$ are faithful. Let $r_Y \in A_N^{**}$ be the ordinary open support*

projection of Y . Then r_Y commutes with p_N , and the support in $\mathcal{A}_N$ of the included corner $(Y^{\sharp_J})^*$ is $p_N r_Y$. Every contractive approximate unit of Y converges weak* in A_N^{**} to r_Y .

If X is one of the boundary pieces used below and ρ_X denotes the ordinary support of $K_X(N)$, while r_X denotes the ordinary support of X , then

$$q_X = p_N r_X = p_N \rho_X. \quad (2)$$

Proof. Let p_ℓ and p_r be DKEP's one-normal support projections in A_N^{**} , for the N - and JNJ -module structures respectively. In the one-normal construction $p_\ell \in N^{**}$ and $p_r \in (JNJ)^{**}$. Since the concrete algebras N and JNJ commute in A_N , their bidual images commute in A_N^{**} : approximate elements of the biduals weak* by bounded nets from the concrete algebras and use separate weak* continuity of multiplication. Hence $p_\ell p_r = p_r p_\ell$.

The one-normal support paragraph preceding DKEP Remark 2.8 identifies $(A_N^{\sharp_N})^* = p_\ell A_N^{**} p_\ell$ and $(A_N^{\sharp_{JNJ}})^* = p_r A_N^{**} p_r$. DKEP Proposition 2.10 identifies the simultaneous normal bidual with their intersection inside A_N^{**} . For commuting projections e, f in a von Neumann algebra, $eRe \cap fRf = (ef)R(ef)$; therefore

$$\mathcal{A}_N = (A_N^{\sharp_J})^* = p_N A_N^{**} p_N, \quad p_N = p_\ell p_r. \quad (3)$$

Apply the same one-normal support statement to Y for each of the two normal structures. The faithfulness hypothesis is exactly the hypothesis needed there for the multiplier actions on Y . Thus r_Y commutes with both p_ℓ and p_r , and the two one-normal biduals of Y are $r_Y p_\ell A_N^{**} p_\ell r_Y$ and $r_Y p_r A_N^{**} p_r r_Y$. Applying Proposition 2.10 to the inclusion $Y \subset A_N$ and using the elementary intersection formula once more gives

$$(Y^{\sharp_J})^* = r_Y p_N A_N^{**} p_N r_Y = (p_N r_Y) \mathcal{A}_N (p_N r_Y).$$

Thus its support projection in $\mathcal{A}_N$ is $p_N r_Y$. The weak* convergence of approximate units to r_Y is the usual open-support property in the ordinary bidual.

It remains to compare X with $K_X(N)$. For the subalgebra boundary pieces used below the faithfulness assumption is automatic: if, for example, $ae_B = 0$, then $E_B(a^*a) = 0$, hence $a = 0$, and the right action is similar; the same holds for $K_X(N)$ since $K(L^2N) \subset X \subset K_X(N)$. By the first part, the normal supports of X and $K_X(N)$ are $p_N r_X$ and $p_N \rho_X$. We show that their simultaneous-normal annihilators coincide.

Let $\phi \in A_N^{\sharp_J}$ vanish on X . DKEP Proposition 3.6 gives $K_X(N) \subset K_X^{\infty,1}(N)$, where $K_X^{\infty,1}(N)$ is the $\|\cdot\|_{\infty,1}$ -closure of X . DKEP Theorem 5.6, applied with the same boundary piece on both sides, says that every $T \in K_X^{\infty,1}(N)$ is killed by every simultaneous-normal functional which vanishes on XA_NX . Since X is hereditary, $XA_NX \subset X$; hence $\phi(T) = 0$ for all $T \in K_X(N)$. The reverse annihilator inclusion is immediate from $X \subset K_X(N)$, which is part of DKEP's construction. Equal preannihilators give the same weak* closed corner in $\mathcal{A}_N$, hence the same support projection. Since q_X is by definition the support of $(K_X(N)^{\sharp_J})^*$, (2) follows. $\square$

For an expected inclusion $B \subset N$, put $H = {}_N L^2 N_B$ and

$$\mathcal{H}_B = H \otimes_B \overline{H} = {}_N L^2 N_B \otimes_B {}_B L^2 N_N.$$

Let $\mathcal{E}_B = \{T_{\eta \otimes_B \bar{\xi}} : \xi, \eta \in H_b\}$.

Lemma 2 (subalgebra fusion). *The boundary piece X_B is hereditarily generated by the positive operators s^*s , $s \in \mathcal{E}_B$, equivalently by $T_\alpha^*T_\alpha$ with α an elementary bounded tensor in $\mathcal{H}_B$. Moreover $\mathcal{H}_B \cong \overline{\mathcal{H}_B}$, and every coefficient operator T_α of a bounded vector in any Hilbert direct sum of copies of $\mathcal{H}_B$ belongs to $K_{X_B}(N)$.*

Proof. DKEP Example 5.11 applied to $H = {}_N L^2 N_B$ says that X_B is the hereditary algebra generated by $T_\xi^* T_\eta$, $\xi, \eta \in H_b$. By (1), $T_{\eta \otimes_B \xi} = T_\xi^* T_\eta = T_\xi^* T_\eta$.

Let Y be the hereditary algebra generated by $\{s^*s : s \in \mathcal{E}_B\}$. Since $\mathcal{E}_B$ is self-adjoint, Y contains s^*s and ss^* . If r is the ordinary support of Y , then $(1-r)s^*s(1-r) = 0$ and $(1-r)ss^*(1-r) = 0$; hence $(1-r)s = s(1-r) = 0$. Thus $s = rsr \in rA_N^{**}r \cap A_N = Y$. So all generators $T_\xi^* T_\eta$ of X_B lie in Y , and the reverse inclusion is immediate.

The conjugation $\xi \otimes_B \bar{\eta} \mapsto \eta \otimes_B \bar{\xi}$ gives $\overline{\mathcal{H}_B} \cong \mathcal{H}_B$. Let $\mathcal{C}$ be the elementary bounded tensors in $\mathcal{H}_B$. They are N - N cyclic, and $T_\alpha \in X_B \subset X_B A_N X_B$ for $\alpha \in \mathcal{C}$. Hence DKEP Theorem 5.10(1) applies, so $\mathcal{H}_B$ is mixing relative to $X_B \times X_B$. For a Hilbert direct sum $\bigoplus_{j \in J} \mathcal{H}_B$, take the vectors of $\mathcal{C}$ placed in one summand. Their N - N span contains the finite-support algebraic direct sum, hence is dense, and the coefficient operator of such a one-summand vector is exactly the coefficient operator above (zero against the other summands). Thus Theorem 5.10(1) holds for the whole direct sum. Theorem 5.10(3), followed by Proposition 5.9, gives $T_\alpha \in K_{X_B, X_B}(N, N) = K_{X_B}(N) A_N K_{X_B}(N) \subset K_{X_B}(N)$ for every bounded vector α . $\square$

Lemma 3 (finite fusion calculus). *Let X be a boundary piece for N , with $N, JN, J \subset M(X)$. Finite Connes fusions, conjugates, and Hilbert direct sums of N - N correspondences which are mixing relative to $X \times X$ are again mixing relative to $X \times X$. Consequently, if K is any such finite-fusion correspondence and $\xi \in K$ is bounded, then $T_\xi \in K_X(N)$.*

For $U \subset V$, put $H_U = L^2 M \otimes_{M_U} L^2 M$. Let $\mathcal{T}_U$ be the linear span of all T_ξ , where ξ is a bounded vector in a finite direct sum of non-empty finite fusions of H_U and $\overline{H_U}$. Then $\mathcal{T}_U$ is a $*$ -algebra, and X_{M_U} has a contractive approximate unit in the norm closure of $\{t^*t : t \in \mathcal{T}_U\}$.

Proof. For a mixing correspondence H , Theorem 5.10 says that

$$\mathcal{C}_H = \{\xi \in H_b : T_\xi \in X A_N X\}$$

is N - N cyclic. Direct sums are handled by placing the cyclic vectors in one summand; their span contains the finite-support algebraic sum and is dense, while the coefficient condition is unchanged. If H and K are mixing and $\xi \in \mathcal{C}_H, \eta \in \mathcal{C}_K$, then $\xi \otimes_N \eta$ is bounded, such elementary tensors are cyclic for $H \otimes_N K$, and

$$T_{\xi \otimes \eta} = T_\eta T_\xi \in (X A_N X)(X A_N X) \subset X A_N X,$$

because X is hereditary. Thus the cyclic-vector criterion proves mixing for fusions. For the conjugate correspondence, the cyclic vectors are $\bar{\xi}$ and $T_{\bar{\xi}} = T_\xi^* \in X A_N X$. Iterating gives the first assertion. Applying Theorem 5.10(3) and Proposition 5.9 to any bounded vector ξ in such a correspondence gives

$$T_\xi \in K_{X, X}(N, N) = K_X(N) A_N K_X(N) \subset K_X(N).$$

The algebra property of $\mathcal{T}_U$ follows term by term from (1) with the order convention $T_\eta T_\xi = T_{\xi \otimes \eta}$: the product of two coefficient operators is the coefficient of the corresponding fused bounded vector, and the adjoint is the coefficient of the conjugate vector. Since $\mathcal{T}_U$ is defined as a linear span, this proves closure under products and adjoints. By Lemma 2, X_{M_U} is hereditarily generated by $T_\xi^* T_\xi$ for elementary tensors in H_U , and these T_ξ 's belong to $\mathcal{T}_U$. If s is a finite positive sum of such generators, then $g_\varepsilon(s) = s(s + \varepsilon)^{-1}$ is in the hereditary algebra generated by the summands. Polynomials in s with zero constant term approximate $g_\varepsilon(s)^{1/2}$, so $g_\varepsilon(s)$ lies in the norm closure of $\{t^* t : t \in \mathcal{T}_U\}$. The usual net over s and $\varepsilon \downarrow 0$ is a contractive approximate unit. $\square$

If $Z \subset U$, then $e_{M_Z} \leq e_{M_U}$; hence $X_{M_Z} \subset X_{M_U}$ and $K_{X_{M_Z}}(M) \subset K_{X_{M_U}}(M)$.

Lemma 4 (intersection of boundary supports). *Let $q_U = q_{X_{M_U}}$. For $U, W \subset V$, with $I = U \cap W$,*

$$0 \leq q_W q_U q_W \leq q_I.$$

Consequently, if $I = \cap_{i=1}^m U_i$, then

$$q_{U_m} q_{U_{m-1}} \cdots q_{U_1} q_{U_2} \cdots q_{U_{m-1}} q_{U_m} \in q_I \mathcal{A}_M q_I.$$

Proof. The graph-product bimodule decomposition and fusion rule of [2, Theorem 5.4 and Proposition 5.5] give

$$H_A \otimes_M H_B \cong \bigoplus_{C \subset A \cap B} H_C^{\oplus m(A, B, C)}. \quad (3)$$

Since $H_A \cong \overline{H_A}$, induction shows that a finite fusion whose labels are $A_1, \dots, A_n$ decomposes into a direct sum of H_C 's with $C \subset A_1 \cap \cdots \cap A_n$. The implementing unitary is M -bimodular, so bounded vectors remain bounded.

Let $\mathcal{P}_R$ be the norm closure of $\{t^* t : t \in \mathcal{T}_R\}$. We claim that

$$a b c \in K_{X_{M_I}}(M) \quad (4)$$

whenever $a, c \in \mathcal{P}_W$ and $b \in \mathcal{P}_U$. It is enough to consider $a = t_1^* t_1$, $b = s^* s$, $c = t_2^* t_2$. Write t_j and s as finite sums of coefficient operators. A summand in the expansion of $t_1^* t_1 s^* s t_2^* t_2$ is

$$T_{\alpha_1}^* T_{\alpha_2} T_{\beta_1}^* T_{\beta_2} T_{\alpha_3}^* T_{\alpha_4},$$

where each α_i is a bounded vector in a finite direct sum of fusions with label W , and each β_i is similarly labelled by U . Replacing T_α^* by $T_{\bar{\alpha}}$ and using $T_\eta T_\xi = T_{\xi \otimes \eta}$, the displayed product is the coefficient operator of

$$\zeta := \alpha_4 \otimes \bar{\alpha}_3 \otimes \beta_2 \otimes \bar{\beta}_1 \otimes \alpha_2 \otimes \bar{\alpha}_1$$

(with finite direct sums distributed over the tensor products). This is a bounded vector in a finite fusion whose label list contains both W and U . By (3), it is carried by an M -bimodular unitary to a bounded vector in a direct sum of H_C 's with $C \subset I$; its coefficient

operator is unchanged under this identification. For each such C , Lemma 2 gives mixing, hence coefficient compactness, relative to X_{M_C} . Since $X_{M_C} \subset X_{M_I}$, the same cyclic vectors are good for X_{M_I} ; the direct-sum argument in Lemma 3 therefore puts the coefficient of the whole direct sum in $K_{X_{M_I}}(M)$. Linearity and norm closure prove (4).

Let $A = \mathbb{B}(L^2 M)$, let $p = p_M$ be the unit of $\mathcal{A}_M$ in A^{**} , let r_R be the ordinary support of X_{M_R} , and let ρ_R be the ordinary support of $K_{X_{M_R}}(M)$. Lemma 1 gives

$$q_R = pr_R = p\rho_R, \quad [p, r_R] = [p, \rho_R] = 0. \quad (5)$$

Choose contractive approximate units $h_\alpha \subset \mathcal{P}_W$ for X_{M_W} and $k_\beta \subset \mathcal{P}_U$ for X_{M_U} . In A^{**} , $h_\alpha \rightarrow r_W$ and $k_\beta \rightarrow r_U$ weak*. By (4), $h_\alpha k_\beta h_\gamma \in K_{X_{M_I}}(M) \subset \rho_I A^{**} \rho_I$. The corner $\rho_I A^{**} \rho_I$ is weak* closed, and multiplication in A^{**} is separately weak* continuous; taking the three successive limits gives

$$r_W r_U r_W \in \rho_I A^{**} \rho_I.$$

This operator is a positive contraction, hence $r_W r_U r_W \leq \rho_I$. Compressing only now to the simultaneous normal bidual, (5) yields

$$q_W q_U q_W = (pr_W p)(pr_U p)(pr_W p) = p r_W r_U r_W p \leq p \rho_I p = q_I.$$

The palindromic assertion follows by induction. If $R_{m-1} \leq q_{\cap_{i < m} U_i}$, then

$$q_{U_m} R_{m-1} q_{U_m} \leq q_{U_m} q_{\cap_{i < m} U_i} q_{U_m} \leq q_{\cap_i U_i},$$

using the two-set case for the last inequality. $\square$

From finitely many relative boundaries to the compact boundary

Lemma 5. *Let N be a separable non-amenable finite factor. Let $X_1, \dots, X_m$ be boundary pieces, $q_i = q_{X_i}$, and suppose*

$$R := q_m q_{m-1} \cdots q_1 q_2 \cdots q_{m-1} q_m \in q_0 \mathcal{A}_N q_0. \quad (6)$$

If N is properly proximal relative to each X_i , then N is properly proximal.

Proof. Assume not. DKEP Lemma 8.5 gives an N -central state ω on $\tilde{S}(N) = \tilde{S}_{\mathbb{K}(L^2 N)}(N)$ with $\omega|_N = \tau$. Fix i and put $q = q_i$. Since q commutes with N and JNJ ,

$$[q^\perp y q^\perp, a] = q^\perp [y, a] q^\perp = 0, \quad a \in JNJ,$$

for $y \in \tilde{S}_{X_i}(N)$; hence $q^\perp y q^\perp \in \tilde{S}(N)$. If $\omega(q^\perp) > 0$, then $y \mapsto \omega(q^\perp y q^\perp) / \omega(q^\perp)$ is an N -central state on $\tilde{S}_{X_i}(N)$. On N_+ it is dominated by $\omega(q^\perp)^{-1} \tau$, hence is normal; its normalization and centrality make its restriction the unique normal trace on the factor N , contradicting relative proper proximality. Thus $\omega(q_i^\perp) = 0$ for all i .

Each q_i , q_0 , and the palindromic product R commute with JNJ , so they lie in $\tilde{S}(N)$; their values below are therefore independent of any extension. Extend ω to a state on the C^* -algebra $\mathcal{A}_N$ only for the following GNS projection argument; the extension is not claimed to be N -central. In its GNS representation, the equality $\omega(q_i^\perp) = 0$ implies q_i fixes the cyclic vector. Hence $\omega(R) = \|(q_1 q_2 \cdots q_m) \xi_\omega\|^2 = 1$. Since R is a positive contraction in $q_0 \mathcal{A}_N q_0$, $R \leq q_0$, and so $\omega(q_0) = 1$.

DKEP Lemma 8.5 provides the canonical N - and JNJ -bimodular u.c.p. map

$$\Theta_0 : \mathbb{B}(L^2 N) \rightarrow q_0 \mathcal{A}_N q_0, \quad \Theta_0(a) = a q_0 \quad (a \in N).$$

Then $\psi(T) = \omega(\Theta_0(T))$ is an N -central state on $\mathbb{B}(L^2 N)$. Since q_0 fixes the GNS cyclic vector, $\psi(a) = \omega(a q_0) = \omega(a) = \tau(a)$ for $a \in N$. This Connes hypertrace contradicts non-amenability of N . $\square$

A relative Bass–Serre paradox

Proposition 1. *Let $N = P *_B Q$ with trace-preserving expectations and $Q = B \overline{\otimes} A$. Assume that A has a trace-zero unitary a , and that P has unitaries w_1, w_2 satisfying*

$$E_B(w_i) = 0, \quad E_B(w_1^* w_2) = 0.$$

Then N is properly proximal relative to X_B .

Proof. Let $H_P = L^2 P \ominus L^2 B$ and $H_Q = L^2 Q \ominus L^2 B$. The AFP normal form is

$$L^2 N = L^2 B \oplus \bigoplus_{\substack{n \geq 1, i_j \in \{P, Q\} \\ i_j \neq i_{j+1}}} H_{i_1} \otimes_B \cdots \otimes_B H_{i_n}.$$

Let p and q project onto the summands whose first letter is P , respectively Q ; then $1 = e_B + p + q$. Right multiplication by B preserves the first letter. Let $x \in P \ominus B$. On $L^2 B$, JxJ creates a first-letter P vector. On a reduced word with first letter Q , right multiplication by x^* either appends or merges at the last letter, but the first letter remains Q and no $L^2 B$ -part is created. On a first-letter P word the first letter remains P , except that a length-one word may produce an $L^2 B$ -component. Hence

$$[p, JxJ] = (1 - e_B) JxJ e_B - e_B JxJ (1 - e_B), \quad [q, JxJ] = 0. \quad (7)$$

The symmetric formula holds with P, p and Q, q interchanged for $x \in Q \ominus B$. Since $E_B(x) = 0$, $e_B JxJ e_B = 0$; hence the two terms in (7) are just right translates of e_B (up to sign), so they lie in $X_B \subset K_{X_B}^{\infty, 1}(N)$.

We now translate DKEP Lemma 6.1. For a fixed operator T , the set of $x \in N$ such that both $[T, JxJ]$ and $[T, Jx^*J]$ lie in $K_{X_B}^{\infty, 1}(N)$ is a von Neumann algebra; this is Lemma 6.1 applied to $B(L^2 N)$ with DKEP's right-action convention $x \cdot T \cdot y = Jy^* J T Jx^* J$. For $T = p$

and $T = q$ this von Neumann algebra contains B , $P \ominus B$, and $Q \ominus B$, hence contains the von Neumann algebra generated by P and Q . Thus $p, q \in S_{X_B}(N)$.

It remains, by DKEP Theorem 6.2, to rule out an N -central state φ on $S_{X_B}(N)$ with normal restriction to N . For $\eta \in qL^2N$, $w_i\eta$ is a reduced vector with first letter P . Moreover $q w_1^* w_2 q = 0$: if a reduced vector begins with Q , left multiplication by the centered P -letter $w_1^* w_2$ produces a vector beginning with P . Thus

$$w_1 q w_1^* + w_2 q w_2^* \leq p.$$

Likewise $a \in Q \ominus B$ and left multiplication by a sends every first-letter P reduced word into a first-letter Q word, so

$$a p a^* \leq q.$$

Since $S_{X_B}(N)$ is an N -bimodule and φ is N -central, these inequalities give $2\varphi(q) \leq \varphi(p) \leq \varphi(q)$, hence $\varphi(p) = \varphi(q) = 0$.

The unitary $h = w_1 a$ is B -Haar: every non-zero power is an alternating reduced word in centered P - and Q -letters, so $E_B(h^n) = 0$ for $n \neq 0$. For $m \neq n$,

$$h^n e_B h^{-n} h^m e_B h^{-m} = h^n e_B h^{m-n} e_B h^{-m} = h^n E_B(h^{m-n}) e_B h^{-m} = 0.$$

The projections $h^n e_B h^{-n}$ are pairwise orthogonal and, by centrality, have the same φ -value; hence $\varphi(e_B) = 0$. This contradicts $1 = \varphi(e_B + p + q)$. By DKEP Theorem 6.2 this is precisely proper proximality relative to X_B (and, in the factor case used below, agrees with the normal-bidual criterion from Lemma 8.5). $\square$

Application to graph products

Fix $v \in V$ and put $L_v = \text{Lk}_\Gamma(v)$. The graph-product reduced-word model, or [3], gives the standard amalgamated free product decomposition

$$M \cong M_{V \setminus \{v\}} *_{M_{L_v}} (M_{L_v} \overline{\otimes} M_v). \quad (8)$$

Let $C(v) = \{r \neq v : r \not\sim_\Gamma v\}$. Since Γ^c is connected, $C(v) \neq \emptyset$. If $C(v)$ contains distinct r, t , choose trace-zero unitaries u_r, u_t and set $w_1 = u_r, w_2 = u_t$. Then $w_i \notin M_{L_v}$, and the reduced-word expectation onto M_{L_v} kills w_i and $w_1^* w_2$. If $C(v) = \{r\}$, connectedness of Γ^c and $|V| \geq 3$ give $s \in L_v$ with $s \not\sim_\Gamma r$; otherwise $\{v, r\}$ would be a component of Γ^c . Set $w_1 = u_r, w_2 = u_s u_r$. Here $E_{M_{L_v}}(w_1) = 0$, $E_{M_{L_v}}(w_2) = u_s E_{M_{L_v}}(u_r) = 0$, and $w_1^* w_2 = u_r^* u_s u_r$ is a reduced word containing the outside vertex r , so its expectation onto M_{L_v} is 0. Thus in all cases

$$E_{M_{L_v}}(w_i) = 0, \quad E_{M_{L_v}}(w_1^* w_2) = 0.$$

A trace-zero unitary in M_v supplies the unitary a in (8), so Proposition 1 shows that M is properly proximal relative to $X_{M_{L_v}}$ for every v .

Enumerate $V = \{v_1, \dots, v_m\}$. Since no vertex belongs to its own link, $\cap_i L_{v_i} = \emptyset$. Lemma 4 puts the associated palindromic product in $q_0 \mathcal{A}_M q_0$. Here $M_\emptyset = \mathbb{C}$, $e_\mathbb{C}$ is the rank-one projection onto $\mathbb{C}\hat{1}$, and DKEP Example 3.4 gives $X_{M_\emptyset} = \mathbb{K}(L^2 M)$.

Finally, CDHJKN Theorem 2.4 implies that a join-irreducible graph product with $|V| \geq 3$ and trace-zero vertex unitaries is a diffuse full factor. A diffuse amenable finite factor is the hyperfinite II_1 factor and is not full, so M is non-amenable. It is separable because the graph is finite and the vertices are separable. Lemma 5, applied to the finite family $\{X_{M_{L_v}} : v \in V\}$, proves that M is properly proximal.

References

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